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Another Technology Race: US-China Quantum Computing Landscape

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There is one assertion on which most American politicians can all agree, at least publicly - China is a dangerous adversary to the United States. To say otherwise or even take a more moderate stance as an elected official, could be perceived as un-American. While there are many narratives one can leverage to justify this position, the development of emerging technologies remains one of the most fiercely contested aspects to the building political tensions between Beijing and Washington. Right now, quantum technologies, more specifically quantum computers (QCs), are maturing in the shadow of the heavily dominated focus on artificial intelligence (AI). Tens of billions of dollars are being funneled into QCs across the world, with China and the United States leading the charge on the development of these next generation computers. In fact, the rising importance of QCs is becoming so prominent that the United Nations (UN) has designated 2025

QCs is becoming so prominent that the United Nations (UN) has designated 2025 as “International Year of Quantum Science and Technology” (IYQ). Similar to other emerging technologies, however, QCs are also becoming another frontier of division, with pathways of cooperation becoming more and more sparse.

In the capital city of China, I spent nearly four weeks talking with quantum scientists and developers in academia, industry and government about what a future with quantum computers might look like. Amidst the backdrop of the growing Great Power Competition between China and the United States, this topic has become a focal point of attention. It was my own curiosity which led me to Peking University, one of China’s most distinguished educational institutions, to ask the Chinese pointed questions about what the reality of this competition looks like from their perspective. My overarching goal was to understand how the rivalry between the United States and China is affecting the development of quantum technologies and where they think this competition is headed. Despite an acceptance to study at the University’s Center on Frontiers of Computing Studies, all of my interviews and exchanges were arranged by means of cold introductions via email and text. While there was approximately a 30% response rate to my inquiries, I was still able to conduct interviews with more than ten Chinese quantum professionals. In the past year at the Geneva Graduate Institute and the Harvard Kennedy School, I have focused on the social aspects of quantum computing, such as governance and policy. My time at Peking University was born out of a pressing desire to analyze the rising tensions between the US and China. Important policy decisions will need to be made based on recommendations from all sectors to cooperate, engage or not engage. My research is intended to exercise considerations, new and old, and present perspectives which need due consideration in policy-making processes.

A Brief Intro to Quantum Computers

Huang Leilei, CEO of Zhengze Quantum Technology Co. Ltd., based in Beijing, believes quantum computers are the inevitable future of a new computing era. She says technologists and developers are reaching the bounds of what is capable with the types of computers we use today, referred to as classical computers. Unlike classical computers, a category which includes everything from the age-old abacus

classical computers, a category which includes everything from the age-old abacus to the smartphone in your pocket to the most advanced modern-day supercomputer on Earth, quantum computers utilize the mysterious properties of quantum mechanics at the sub-atomic level. Through quantum mechanical phenomena, such as entanglement and superposition, QCs are poised to exponentially increase computing efficiency and help solve currently intractable problems. In short, this means that QCs will be capable of performing calculations that are fundamentally different and physically not possible for any existing classical computer. These new machines are believed to have the ability to revolutionize many tasks across all sectors and industries and unlock newfound capabilities. Applications are expected to be substantial for financial firms, medical research efforts, logistics and supply chains, digital security and cryptography and so much more. For many in this field there is much excitement, albeit mixed with speculation, as to what these revolutionary computers will be capable of doing. Only further research and development (R&D) can corroborate this excitement.

The US-China Divide

Rapid developments, and the associated hype, of leading-edge technologies introduces, yet again, new understandings of what it means to be alive in the year 2025. Many of these technologies matriculate in society as both general purpose and dual-use in nature. According to the Institute for Economics and Peace, 2024 exemplified the 12th consecutive year of declining peace across the world, listing new technologies as a contributing factor to this trend. Terms like “New Cold War”, “Great Power Competition” and Graham Allison’s “Thucydides Trap” have placed the two leading superpowers of the world in a neck-and-neck race for technological supremacy for fear of inevitable suppression or even possible destruction of the other.

Amidst this sensitivity, walking into a Chinese center for leading technology and introducing myself as an American graduate student focusing on quantum information and science technology (QIST) policy, definitely captured the attention of many I hoped to interview. Prior to arriving in China, the VISA application process presented surprisingly no resistance in obtaining permission to conduct

research in Beijing. My VISA application paperwork included an acceptance letter to Peking University's Center on Frontiers of Computing Studies with the stated goal to research topics in the field of quantum information and science technologies and also listed my prior military service as a pilot trained in electronic warfare. However, on the flip side, the same VISA experience I had is not necessarily reflected for Chinese nationals hoping to study in the US. According to many of the individuals I interviewed, they mentioned that from their personal experiences, Chinese nationals who want to travel to the United States, but are associated with the field of quantum technologies, are unlikely to receive permission to enter. A scientist at Peking University did mention that institution of origin and purpose of visit do play a role in the probability of receiving permission to enter the United States. For example, quantum computing researchers from Tsinghua University, another prestigious institution in Beijing, have more difficulty than researchers at other universities due to its reputation of having technical expertise. While entering the US as a Chinese national with the purpose of attending a quantum conference is possible, I was told one story of a Chinese scientist who received permission to travel to America but faced hours of interrogation upon landing in the United States. While I cannot verify this particular story, former US ambassador to China, Nicholas Burns, recently confirmed in a Foreign Affairs podcast that if Chinese students trying the come to the US are identified as being involved in a science, technology, engineering or mathematics (STEM) field and have the potential to help the People's Liberation Army (PLA) or other fields of Chinese national interest, that the US government is going to turn those students away. Ironically, I thought of this as a sort of import control of human capital, akin to the technology export controls that have been placed on China.

Friction in the allowance for exchanges of quantum discoveries continues to have a developmental impact for both sides. For quantum experimentalists and practitioners, the practice of US-China collaboration is falling to the wayside. In 2023, the Office of Investment Security within the Department of the Treasury delineated prohibitions on investments as they relate to quantum technologies. Late last year, a Mercator Institute for China Studies (MERICS) report referenced the US's speculation that the quantum industry could begin to closely reflect the

semiconductor industry as it relates to US and China competition. But for academics and theorists, despite QIST having become a sensitive matter closely associated with national security, there is still room to learn from each other. Most of this cooperative learning may happen through reading publicly published papers or connecting with former colleagues. In years past, many Chinese quantum scientists were educated in the United States and to this day have maintained these connections with their former colleagues; this was predominantly before technology export controls began to take effect in 2018. Dr. Xiao Yuan, an assistant professor at Peking University and who previously worked as a post doc at Stanford University in the field of quantum computing, worries for a day when even academics and theorists are not able to collaborate. He believes that if public papers start being recalled, or not published at all, then the relationship will find itself in a new stage of heightened paranoia that will not bode well for either party.

In reflex, Dr. Abolfazl Bayat, head of the Physics of Quantum Information Technologies group within the Institute of Fundamental and Frontier Sciences at the University of Electronic Science and Technology of China, said that the University has started to actively discourage their own students from applying for a VISA to the US, let alone actually making the journey. Bilateral cooperation between China and the US is continuing to morph into a foreign concept for both.

Export Controls

Since 2018, the US has imposed export controls on the transfer of technologies to China, heavily targeting semiconductors and other technologies dubbed relevant to the ability of the Chinese Communist Party (CCP) to keep pace with the US's AI dominance. Doubling down in 2022, export controls continued to intensify to restrict China's technology and military modernization. These export controls now also include restrictions on China's ability to access components for building quantum computers and other QIST hardware.

For most Chinese university researchers in this field, the current export controls related to QIST are not necessarily targeted towards, nor have a direct impact on their work – part of the reason why US-China quantum collaboration at the academic level remains alive. While it is yet to be seen how effective quantum

related export controls will be, academics focusing on theoretical studies often do not require sought after hardware components which are becoming increasingly difficult for industry to acquire. There was mention, however, that Qiskit, IBM's software stack used for quantum algorithm development, was no longer available to users in China. Chinese academics at Peking University expressed that they see value in using the Qiskit software stack and have been able to use virtual private networks (VPNs) to circumvent this restriction. It is yet another confirmation to the narrowing scope of free cooperation. As for the government representatives I spoke to within the Chinese Ministry of Science and Technology, they expressed less concern than academic and industry professionals with regard to the export controls coming from the US Department of Commerce. While they do admit that these control measures are having a small impact on their development timelines, and while they wish that they were not being implemented, they are still successfully navigating through these obstacles. They say the export controls are only a small hurdle and out of necessity, China is being forced to innovate more vigorously; eventually it will become far less dependent on foreign suppliers. There is a level of pride in this fact. And it is true across several technologies, not just quantum computing. The recent announcement of DeepSeek serves as a blaring example.

The majority of Chinese quantum physicists and industry professionals I spoke with highlighted that the "small yard, high fence" tactics and less-than-sophisticated foreign policy tools, like export controls, are somewhat puzzling to them at this stage of the field of quantum computing. While quantum sensing and quantum communications, two sub-fields of QIST, have started to realize real-world applications, quantum computing, the third and most consequential sub-category of QIST, is still far from being at a level of commercialization. At this early stage, quantum scientists are still theorizing and testing several techniques to achieve an effective quantum computer which can execute the tasks it is intended to perform. A key complication to realizing what scientists call a scalable and fault-tolerant quantum computer (SFTQC) is a concept called error-correction. The sensitive and precarious nature of manipulating particles at a sub-atomic scale presents many challenges caused by different types of interference which result in "errors". Of the development methods being tested, no singular technique has yet

been identified as a clear winner.

Many Chinese are wondering why the field has become so pointed and secretive between the two governments when there is still so much work to be accomplished before any substantive threat can be realized. Of course, an argument can be easily crafted from either side that any lead, no matter how small, in the development of a consequential technology is in the interest of safeguarding national security interests. But, of note, national security was one of the least cited objectives to come from quantum computer development throughout my conversations with the Chinese. They expressed far more interest, at least in what was conveyed to me, in the societal benefits of QCs. To no surprise, competition, as it relates to national security, is well understood in the eyes of the Chinese; but many are viewing QCs as a technology that holds far more applications than just a tool of technological warfare. Of those I interviewed, they believe quantum development can be more efficiently progressed if each nation wasn't developing quantum computers in their own silos. But in my opinion, there is a caveat to this sentiment. I did not speak to a single Chinese quantum stakeholder who did not unanimously agree that the United States is currently leading the quantum computing charge by at least an estimated 3-5 years. Perhaps, if China felt they were in the lead, there would be a keener understanding to wanting to maintain the headway they had, no matter the stage of development. Still, Jian Chen, CEO of Little Quantum, says this competition is only delaying a very important technology which has the ability to address some of humanity's toughest challenges. Quantum simulation, for example, is one application of a quantum computer which allows the manipulation of atomic structures to develop novel advancements in biomedicine and materials science. Chen compared the diplomatic barriers to a more kinetic analogy. He told me "In a sense, killing 1,000 enemies but losing 800 of your own is actually affecting both sides. In ten years, twenty years, or fifty years, it will not necessarily matter who had the smaller impact and who had the greater impact on each other." The latter half of this statement, perhaps, minimizes the significance of the security implications quantum computers seem to pose. However, it is another testament to the above comment that the Chinese view this technology as too important for societal benefits to not find ways to accelerate QCs together, rather than find ways to hold each other back.

Quantum Cooperation

Technological competition inevitably involves considerations beyond the sole initial discovery of a new hardware, software or capability. If one is to compete on a global scale in innovation, one needs to have competitive regulation as well. Throughout history, global standards and regulations have frequently been required to provide some common level of protection against the dangers of a new technology. Multilateral international organizations have provided these frameworks in the past such as the United Nations (UN), World Trade Organization (WTO) and International Telecommunications Union (ITU). The United States and China do not have robust bilateral treaties which allow the two superpowers to keep each other in check as their capabilities surpass much of the rest of the world.

Several people I interviewed expressed that they are seeing serious problems with the lack of US-China partnership and cooperation. They believe there are certain issues which cannot be solved through one country having more scientists in the lab or more dollars in the bank. Some problems must rely on good old-fashioned diplomacy and the understanding that sometimes all of humanity faces a universal danger. Huang Leilei believes that there are still enough unknowns about quantum computers and their applications that they remain a potential threat. For example, aside from the cryptographic dangers QCs bring into play, Huang Leilei says there are many scientists who hypothesize that the human brain naturally operates through quantum physical properties. Referencing the common artificial intelligence doomsday scenario where computers become advanced enough to "take over", she says there are some who worry mature quantum computers, used in conjunction with AI, may allow for computer consciousness. The diplomacy associated with addressing a common enemy, whether it be the next global pandemic or a new doomsday machine, is an area of cooperation which can humanize the shared wants and needs of the other side. The US-China efforts on the fentanyl crisis serve as one example of a cooperative framework which can be adapted elsewhere.

Authoritarianism vs. Democracy

Authoritarianism vs. Democracy

On this matter, the question does arise as to whether the differences between an authoritarian state and a democratic one will have a significant impact on the ability to cooperate on governance and regulation while this competition continues to rage. Methods of funding, for instance, will impact development roadmaps and areas of concentration. In China, the money comes from the top levels of government down to the point of investment. This places pressure on top quantum thought leaders to be high performers to attract attention and thereby receive state funding. On the other hand, while the United States does have quantum government initiatives and investments, the private sector leads the charge in funding and research and development. The number of startups and heavy investment coming from venture capitalists and private enterprises makes it possible for quantum development to derive from a wider array of sources. In China, Dr. Bayat mentioned, venture capitalists have a shorter time threshold for results, saying that "VCs want to see a return on investment in five years or less. Development of effective quantum computers is not going to happen in 5 years."

China's lack of private funding will likely affect its ability to keep up with the West. China's government investment into quantum information technologies is reported to be approximately $15 billion. Compared to other developed nations, with numbers hovering between $2-3 billion, China's government investment dwarfs all other public sector quantum investments from governments around the world. Despite this apparent public funding discrepancy, a 2022 RAND Corporation report found that China's private sector funding is only 3% of US corporate and venture capital (VC) funding. What's more is that there are many Chinese who are even skeptical of China's publicized public sector funding amount. Dr. Chao-Yang Lu, a famous quantum physicist at the University of Science and Technology of China (USTC) and close associate of Dr. Jian-Wei Pan, China's "Father of Quantum Computing", believes the actual number is much smaller than $15 billion, perhaps only one-third of the advertised amount. But he believes a benefit of China's top-down government funding for quantum researchers is that they are able to receive financial support to work on the projects they believe in most. Dr. Chao-Yang Lu says that regardless of an authoritarian or democratic governance framework, and regardless of government funding strategies, the ability to succeed in this quantum

race will rely on which country can attract the most talent and offer the most research freedom. Dr. Abolfazl Bayat, on the other hand, believes this funding strategy leads to large inefficiencies in their quantum spending. Dr. Bayat recognizes that the US and other national quantum strategies require robust, defined roadmaps and strategies to receive more targeted funding, rather than dispersing funds as liberally as China does.

It was mentioned in a number of discussions that some of this inefficiency has led China to develop a sharper talent in finding broader applications for existing technologies than their ability to keep up with the development of breakthrough technologies. Jeffrey Ding, a professor of political science at George Washington University, talks about the importance of the diffusion of already developed general-purpose technologies within a society possibly outweighing the benefits of the development of leading sector technologies. There is great value in the fluency and proficiency of finding many new applications for existing technologies. But perhaps countries who embrace this method risk falling behind on next generation capabilities. It would appear to me that China is splitting resources and trying to invest in both routes. However, I am skeptical of their ability to succeed at both in the short term due to the capital-intensive nature of competing in so many fronts of emerging technologies.

The CCP's regulatory crackdown in 2020 has also had a large impact on China's technology sector. Xi Jinping and the communist party enacted harsh antitrust regulations against big tech companies for engaging in monopolistic behaviors. This resulted in a retreat of investment and a tough landscape for big and small firms to stabilize their businesses. While much of this has recently reversed course, many companies are still facing hardships because of this regulatory fallout. Dr. Bayat says many Chinese entrepreneurs are turned off by the idea of innovating and building businesses in China and would much rather travel to nearby countries like Singapore or Vietnam where they have greater promise for building fruitful enterprises. In the US, the startup environment is flourishing at levels not comparable to China.

Bottom Line

The quantum race is not unique in that almost all aspects of the US-China landscape currently seem to be at odds. For the past several years the world seems to be at the vergence of a new geopolitical era. The tides of a unipolar world order shifting into a multipolar one seems more plausible with each passing year. Other countries are competing for dominance in their respective domains and China is no exception. They have undoubtedly put on an impressive performance economically, technologically and militarily in the previous two decades and this is unsettling to many in the West. Change introduces uncertainty and the incumbent power is feeling threatened. As we have entered into the Fourth Industrial Revolution, technological dominance will equate to dominance in most other fields. This mindset has given rise to a “New Cold War” narrative. Sharp and pointed political language and its diffusion through American society is fortifying the divide in the US-China relationship. In an exchange with Chris Miller, semiconductor expert and author of the best-seller “Chip War”, I asked Miller why he chose to use the word “war” in the title of his work. Miller responded, “I think competition is the reality. Those analysts who pretend it isn't aren't making it less real, they're just sticking their head in the sand. I wouldn't say that readers in Asia were any more or less skeptical of the title. If anything, I think readers in East Asia were more cognizant of the competitive realities than American readers.”

Intense rivalry to the existing order is an inevitable truth of present-day geopolitics. It would be naive to not recognize it as such. However, narratives that war is inescapable, similar to increasing references to the Thucydides Trap, should serve more as cautionary tales of actions that need to be averted rather than perceived as preordained collisions. Competition in the field of emerging tech is not a new development and it is expected. But competition should not be justified with the reasoning that every interaction come from an adversarial stance. Coexistence between the two superpowers is going to be a defining characteristic of twenty-first century geopolitics. Coexistence implies room for both competition and cooperation. Competition can be healthy and encourage innovation to be at its sharpest. But competition does not mean we can afford to mobilize entire populations to adopt an enemy mindset and dangerous attitudes towards our competitors. This has a foul tendency to dehumanize each other and entice a self-fulfilling prophecy. There are realities and justifications to understanding

components of complex relationships as adversarial, but to apply this perspective at large makes us as dangerous to our competition as they see us.

Quantum computing has promising applications to help change the world in ways we still cannot fully foresee. There is broad recognition of the dangers it presents to digital security, and these should be carefully considered. But importantly, there are likely pathways to engage in cooperative development of QCs while still prioritizing the safeguarding of intellectual property and preserving national security. Perhaps, cooperative opportunities may exist for industry and private enterprises in fields such as chemistry, biomedicine, multi-sector optimizations and much more. This is an opportune time in the early-stage development to identify these areas. Non-engagements in cooperation should persist in fields of security, military and defense applications. This will keep competition, which invigorates innovation, alive. Restrictions in these national security applications serve to buttress the importance of recognizing international law and respecting sovereignty.

Quantum diplomacy is an opportunity to act on a technology that nobody has yet developed. It is a chance to steer away from current tides of the growing divide between the United States and China to more calm waters. We are witnessing ever-rising tensions as the race for AI supremacy plays out before us. It will take concerted diplomatic efforts to rewrite this narrative for the technologies of tomorrow. As national quantum strategies continue to emerge throughout the world, pathways of cooperation need to be identified and enshrined. Whether this cooperation manifests itself through regulations on quantum hardware, quantum software, academic partnerships, government communications, bilateral treaties, new trade agreements or any number of other diplomatic tools, the time to lay the foundation of renewed technological cooperation is upon us.

Brian Moscioni is a MC/MPA 2025 candidate at The Harvard Kennedy School. He is pursuing a dual-degree with the Geneva Graduate Institute for a master's degree in international development. Brian has focused his studies on emerging technology policy and governance.

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